

# Advanced Data Analytics 2025

## Course Wrap-Up and Outlook

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Week 14 – Monday, December 15, 2025

# Today's Agenda (120 minutes)

## Part I: The Journey (40 min)

- From Newbie to AI Pioneer
- Course roadmap review
- The Machine Learning landscape

## Part II: Core Concepts (40 min)

- Supervised Learning recap
- Deep Learning highlights
- Probabilistic methods
- Unsupervised & RL

## Part III: Integration (20 min)

- Connecting the dots
- Cross-cutting themes
- Method selection guide

## Part IV: Looking Ahead (20 min)

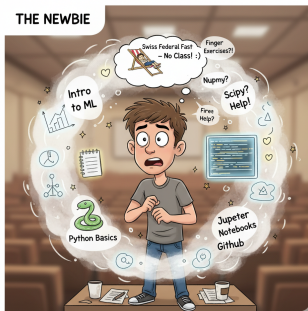
- Current trends in ML/AI
- Career paths
- Capstone project advice
- Final thoughts

# Part I

The Journey: From Newbie to AI Pioneer

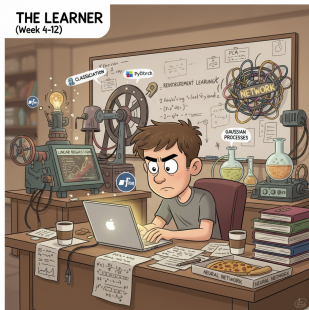
# Your Transformation Over 14 Weeks

## Weeks 1-3 *The Newbie*



Python? NumPy? Help!

## Weeks 4-12 *The Learner*



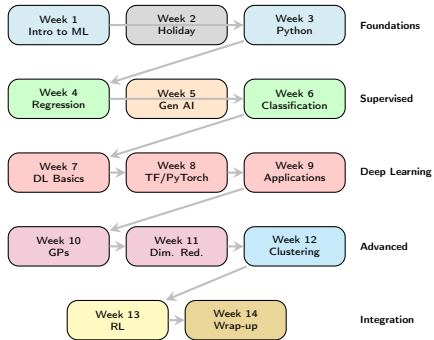
Neural networks, GPs,  
Clustering...

## Week 14+ *The AI Pioneer*

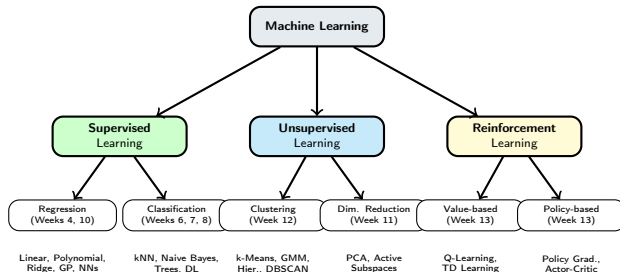


Ready to tackle real problems!

# The Course Roadmap



# The Machine Learning Landscape



# What You've Learned: By the Numbers

## Methods Covered

- 7+ regression techniques
- 5+ classification algorithms
- 4 clustering methods
- 3 dimensionality reduction approaches
- 2 reinforcement learning paradigms

## Tools & Frameworks

- Python ecosystem
- NumPy, SciPy, Pandas
- scikit-learn
- TensorFlow & PyTorch
- GFlow
- Jupyter Notebooks

## Application Domains

Finance (option pricing, portfolio optimization), Economics (dynamic models), Climate science, Natural language processing, Computer vision, Time series forecasting

# Part II

## Core Concepts Recap



# Supervised Learning: The Big Picture

## The Setup

Given training data  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ , learn a function  $f : \mathcal{X} \rightarrow \mathcal{Y}$

### Regression ( $y \in \mathbb{R}$ )

- Linear:  $f(x) = w^\top x$
- Polynomial:  $f(x) = \sum_j w_j \phi_j(x)$
- Neural Networks:  $f = f_L \circ \dots \circ f_1$
- Gaussian Processes

### Classification ( $y \in \{0, 1, \dots, K\}$ )

- k-Nearest Neighbors
- Naive Bayes
- Decision Trees
- Neural Networks (softmax)

## Key Principle

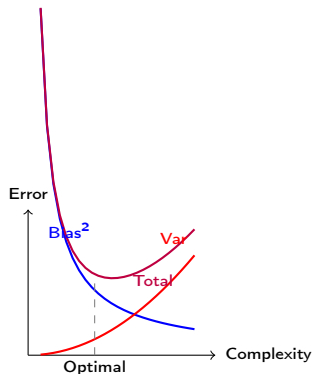
Minimize empirical risk while controlling model complexity

# The Bias-Variance Tradeoff

## Expected Error Decomposition

$$\mathbb{E} \left[ (y - \hat{f})^2 \right] = \text{Bias}^2 + \text{Var} + \sigma^2$$

- **High Bias:** Too simple → **Underfitting**
- **High Variance:** Too complex → **Overfitting**
- **Goal:** Find the sweet spot!



**Strategies:** Regularization, Cross-validation, Early stopping, Dropout, Ensembles

# Gradient Descent: The Workhorse of ML

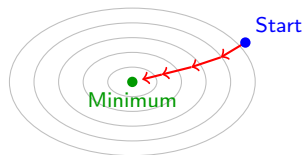
## The Algorithm

**Goal:** Minimize loss  $\mathcal{L}(\theta)$

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_{\theta} \mathcal{L}(\theta^{(t)})$$

**Variants covered:**

- **Batch GD:** Use all data each step
- **Stochastic GD:** Use one sample
- **Mini-batch GD:** Use small batches
- **Adam, RMSprop:** Adaptive learning rates

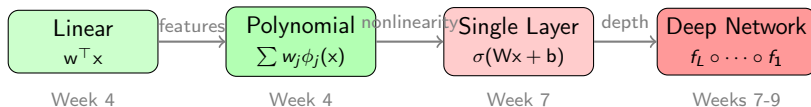


Loss landscape

## Key Insight

SGD's noise actually helps escape local minima – crucial for deep learning!

# Linear Regression to Neural Networks: A Continuum



**Common thread:** Minimize a loss function

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, f(x_i; \theta))$$

**Regularization throughout:**

- Ridge/Lasso for linear models
- Weight decay for neural networks
- Dropout, batch normalization

# Classification: From Simple to Sophisticated

## k-Nearest Neighbors

- Non-parametric, instance-based
- Prediction: majority vote of k nearest
- Key: choice of k and distance metric

## Naive Bayes

- Probabilistic:  $P(y|x) \propto P(y) \prod_j P(x_j|y)$
- “Naive” independence assumption
- Fast, works well for text

## Decision Trees

- Interpretable, hierarchical rules
- Split on features to maximize info gain
- Prone to overfitting → pruning

## Ensemble Methods

- **Bagging**: Average many trees (Random Forest)
- **Boosting**: Sequentially fix errors
- Often best “out of the box” performance

## Evaluation Metrics

Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

## Feed-Forward Networks (FFN)

- Layers: Input  $\rightarrow$  Hidden  $\rightarrow$  Output
- Universal approximation theorem
- Applications: tabular data, function approximation

## Recurrent Neural Networks (RNN)

- Sequential data processing
- Hidden state carries information
- Challenge: vanishing gradients

## Long Short-Term Memory (LSTM)

- Gates: forget, input, output
- Cell state for long-term memory
- Applications: time series, NLP

## Transformers (Glimpse)

- Self-attention mechanism
- Parallel processing
- Foundation for LLMs (GPT, etc.)

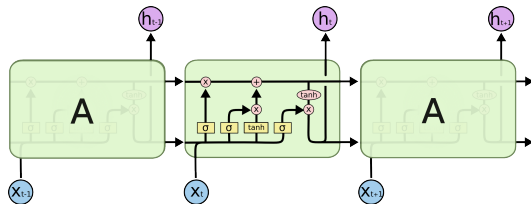
# LSTM: The Gates that Remember

## Three Gates + Cell State:

- 1 **Forget:**  $f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$
- 2 **Input:**  $i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$
- 3 **Output:**  $o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$

**Cell State:**  $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$

**Applications:** Stock prices, weather, time series



Source: C. Olah's blog

# TensorFlow vs PyTorch: Your Deep Learning Toolbox

## TensorFlow

- Production-ready, scalable
- TensorBoard for visualization
- Keras API for ease of use
- Static graph (eager mode available)
- Strong deployment ecosystem

## PyTorch

- Research-friendly, Pythonic
- Dynamic computation graph
- Easy debugging
- Strong in academia
- torchvision, torchaudio, etc.

## Practical Advice

- Both are excellent choices – pick based on your use case
- For research/prototyping: PyTorch often preferred
- For production/deployment: TensorFlow has mature tools
- Learn one well, the concepts transfer!



# Neural Networks for Economics & Finance

Applications covered in Weeks 9 & 14:

## Physics-Informed NNs (PINNs)

- Embed PDEs as constraints
- Option pricing (Black-Scholes)
- Respects domain knowledge

## Deep Equilibrium Networks

- Solve dynamic economic models
- Handle high-dimensional states
- Uncertainty quantification

## Key Insight

Neural networks as **function approximators** for:

- Value functions
- Policy functions
- Pricing functions
- Equilibrium mappings

**Why it works:**

- Universal approximation
- Handles curse of dimensionality
- GPU acceleration

# Gaussian Processes: Bayesian Non-parametric Regression

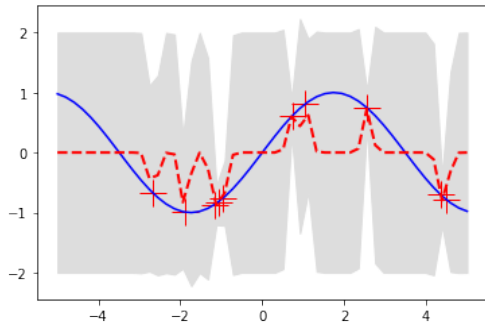
## Key Idea

Prior over **functions**:  $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$

## Components:

- Mean function  $m(x)$
- Kernel  $k(x, x')$  (RBF, Matérn)

**Posterior:**  $f_*|X, y \sim \mathcal{N}(\mu_*, \Sigma_*)$



GP prediction with uncertainty

## Why GPs Matter

**Built-in uncertainty quantification!** Know what you don't know.

# The Curse of Dimensionality & Solutions

## The Problem

As dimension  $d$  increases:

- Data becomes sparse
- Distance metrics lose meaning
- Sample complexity explodes
- $N \sim O(\epsilon^{-d})$  for  $\epsilon$ -covering

## Solutions (Week 11)

- **PCA:** Linear projection to lower-D
- **Active Subspaces:** Find important directions
- **Deep Kernel Learning:** Non-linear dim. reduction + GP
- **Active Learning:** Query where it matters

## Principal Component Analysis:

- Find orthogonal directions of maximum variance
- Project data:  $z = W^T(x - \mu)$
- Keep top  $k$  components explaining most variance

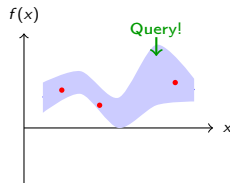
## The Idea

Sample where **uncertainty is highest!**

### Acquisition Functions:

- **Uncertainty:**  $x^* = \arg \max_x \sigma(x)$
- **Expected Improvement**
- **Information Gain**

**Benefits:** Fewer samples, targeted exploration



# Clustering: Finding Structure in Data

## k-Means

- Assign points to nearest centroid
- Update centroids iteratively
- Converges to local minimum
- Choose  $k$  via elbow method, silhouette

## Gaussian Mixture Models

- Soft clustering (probabilities)
- EM algorithm for fitting
- More flexible than k-means

## Hierarchical Clustering

- Agglomerative (bottom-up)
- Divisive (top-down)
- Dendrogram visualization
- No need to specify  $k$  upfront

## DBSCAN

- Density-based
- Finds arbitrary shapes
- Handles noise/outliers
- Parameters:  $\epsilon$ , minPts

# Reinforcement Learning: Learning from Interaction

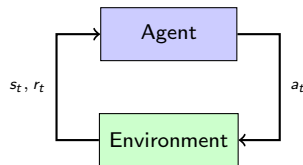
## The Setup

Agent interacts with environment:

- **State**  $s_t$ : Current situation
- **Action**  $a_t$ : Agent's choice
- **Reward**  $r_t$ : Feedback signal
- **Policy**  $\pi(a|s)$ : Decision rule

**Goal:** Maximize  $V^\pi(s) = \mathbb{E} [\sum_{t=0}^{\infty} \gamma^t r_t | s_0 = s]$

**Q-Learning:**  $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$



# RL Application: Portfolio Optimization

## Problem Formulation:

- **State:** Portfolio weights, market conditions
- **Action:** Rebalancing decisions
- **Reward:** Risk-adjusted returns

## Why RL for Finance?

- Sequential decision problem
- Non-stationary markets
- Complex constraints
- Transaction costs matter

### Markowitz meets RL

- Classic: One-period optimization
- RL: Multi-period, dynamic
- Learn optimal policy over time
- Handle realistic constraints

Covered in Week 13

Least-squares policy iteration for portfolio choice with realistic market dynamics

# Part III

## Connecting the Dots



# Cross-Cutting Themes

## 1. Optimization Everywhere

- Linear regression → least squares
- Neural networks → SGD
- k-means → coordinate descent
- RL → policy optimization

## 3. Probabilistic Thinking

- Naive Bayes
- Gaussian Processes
- Mixture Models
- Uncertainty quantification

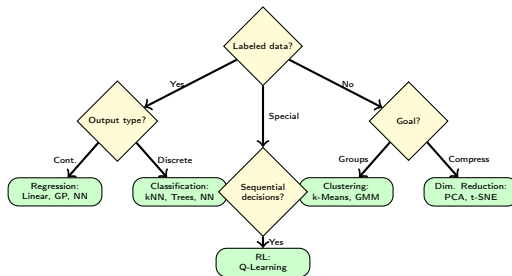
## 2. Regularization

- Ridge, Lasso (linear)
- Weight decay, dropout (DL)
- Kernel choice (GPs)
- Tree pruning (decision trees)

## 4. Feature Representation

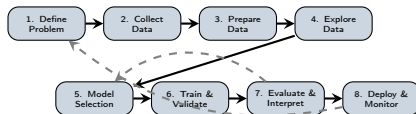
- Feature engineering (manual)
- Kernel methods (implicit)
- Deep learning (learned)
- PCA (linear extraction)

# Method Selection Guide



**Reality:** Often combine methods! GP + Deep Learning, Clustering + Classification, etc.

# The Data Science Workflow



**This course focused on:**

- Steps 5-7 (modeling)
- Understanding algorithms
- Proper evaluation

**In practice, also critical:**

- Data quality & preprocessing
- Feature engineering
- Deployment challenges

# Common Pitfalls to Avoid

## Data Leakage

- Using test data for training
- Feature engineering on full dataset
- **Solution:** Strict train/test split

## Selection Bias

- Non-representative data
- Survivorship bias
- **Solution:** Careful data collection

## Overfitting

- Complex model, limited data
- Perfect training, poor test
- **Solution:** Regularization, cross-validation

## Ignoring Uncertainty

- Point predictions only
- Overconfident models
- **Solution:** Use GPs, ensembles, Bayesian methods

# Part IV

## Looking Ahead

# Current Trends in ML/AI (2025)

## Large Language Models

- GPT-4, Claude, Llama, Gemini
- Transformers at scale
- Prompt engineering
- Multimodal capabilities

## Foundation Models

- Pre-train once, fine-tune many
- Transfer learning at scale
- Vision, language, multimodal

## Scientific ML

- Physics-informed NNs
- Neural operators (PDEs)
- Drug discovery, materials
- Climate modeling

## Responsible AI

- Fairness & bias mitigation
- Explainability (XAI)
- Privacy-preserving ML
- AI safety research

# Career Paths in Data Science / ML

## Industry Roles

- **Data Scientist:** Analysis + ML
- **ML Engineer:** Production systems
- **Research Scientist:** New methods
- **Quant:** Finance + ML
- **AI Product Manager:** Strategy

## Skill Development

- **Deepen:** Choose a specialization
- **Broaden:** Software engineering skills
- **Practice:** Kaggle, personal projects
- **Network:** Conferences, meetups
- **Stay current:** Papers, blogs, courses

## Key Differentiators

- Domain expertise (finance, healthcare, climate...)
- Communication skills (explain ML to non-experts)
- End-to-end project experience (your capstone!)

# Capstone Project: Final Advice

## Structure (reminder):

- 1 Abstract
- 2 Introduction
- 3 Research question & literature
- 4 Methodology
- 5 Data description
- 6 Implementation
- 7 Results
- 8 Conclusion

## Tips for Success:

- Start with a clear question
- Use methods appropriate to your data
- Show understanding, not just results
- Discuss limitations honestly
- Clean, documented code
- Practice your video presentation

## Deadline

**Sunday, January 11th, 2025, 23:59**

Submit to: Maria Pia Lombardo



# Resources for Continued Learning

## Books:

- “An Introduction to Statistical Learning” – free online!
- “Deep Learning” (Goodfellow et al.) – free online!
- “Pattern Recognition and ML” (Bishop)
- “Reinforcement Learning” (Sutton & Barto) – free online!

## Online Courses:

- Stanford CS229, CS231n, CS224n
- fast.ai (practical DL)
- Coursera ML specializations

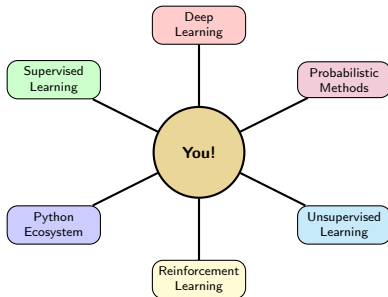
## Stay Updated:

- arXiv (cs.LG, stat.ML)
- Papers with Code
- Distill.pub (visual explanations)
- AI research blogs (OpenAI, DeepMind, Google AI)

## Practice:

- Kaggle competitions
- GitHub projects
- Replicate papers
- Build your portfolio

# Summary: Your ML Toolkit



# Thank You!

## Contact Information

### Instructor:

Simon Scheidegger  
simon.scheidegger@unil.ch

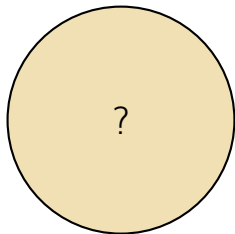
### Teaching Assistants:

Maria Pia Lombardo  
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Good luck with your capstone projects!

## Questions & Discussion



# Appendix

## Reference Slides

## Quick Reference: Loss Functions

| Task           | Loss Function | Formula                                  |
|----------------|---------------|------------------------------------------|
| Regression     | MSE           | $\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2$ |
| Regression     | MAE           | $\frac{1}{N} \sum_i  y_i - \hat{y}_i $   |
| Classification | Cross-entropy | $-\sum_i y_i \log(\hat{y}_i)$            |
| Classification | Hinge (SVM)   | $\max(0, 1 - y_i \hat{y}_i)$             |

### With Regularization:

- Ridge (L2):  $\mathcal{L} + \lambda \|w\|_2^2$
- Lasso (L1):  $\mathcal{L} + \lambda \|w\|_1$
- Elastic Net:  $\mathcal{L} + \lambda_1 \|w\|_1 + \lambda_2 \|w\|_2^2$

# Quick Reference: Evaluation Metrics

## Regression:

- MSE:  $\frac{1}{N} \sum (y - \hat{y})^2$
- RMSE:  $\sqrt{\text{MSE}}$
- MAE:  $\frac{1}{N} \sum |y - \hat{y}|$
- $R^2$ :  $1 - \frac{SS_{res}}{SS_{tot}}$

## Clustering:

- Silhouette score: Measure cluster cohesion vs separation
- Davies-Bouldin index: Lower is better
- Adjusted Rand Index: Compare to ground truth

## Classification:

- Accuracy:  $\frac{TP+TN}{Total}$
- Precision:  $\frac{TP}{TP+FP}$
- Recall:  $\frac{TP}{TP+FN}$
- F1:  $2 \cdot \frac{P \cdot R}{P+R}$
- AUC-ROC

## Quick Reference: Key Equations

Linear Regression (closed form):

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Ridge Regression:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

Logistic Regression:

$$P(y = 1|x) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

Softmax (multi-class):

$$P(y = k|x) = \frac{e^{\mathbf{w}_k^\top \mathbf{x}}}{\sum_j e^{\mathbf{w}_j^\top \mathbf{x}}}$$

GP Posterior Mean:

$$\mu_* = \mathbf{K}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$$



# Week-by-Week Quick Reference

| Week | Topic          | Key Methods                    | Tools                  |
|------|----------------|--------------------------------|------------------------|
| 1    | Intro to ML    | –                              | NuVolos setup          |
| 2    | Holiday        | –                              | –                      |
| 3    | Python         | –                              | NumPy, Pandas, Jupyter |
| 4    | Regression     | Linear, Polynomial, Ridge, SGD | scikit-learn           |
| 5    | Gen AI         | LLMs, Agents                   | (Guest lecture)        |
| 6    | Classification | kNN, Naive Bayes, Trees        | scikit-learn           |
| 7    | Deep Learning  | MLP, Backprop                  | TensorFlow             |
| 8    | DL Frameworks  | RNN, LSTM                      | TF, PyTorch            |
| 9    | Applications   | PINNs, DEQN, Recap             | –                      |
| 10   | GPs            | GP Regression, Kernels         | GPFlow                 |
| 11   | Dim. Reduction | PCA, Active Subspaces          | scikit-learn           |
| 12   | Clustering     | k-Means, GMM, DBSCAN           | scikit-learn           |
| 13   | RL             | Q-Learning, Portfolio          | –                      |
| 14   | Wrap-up        | Integration, Outlook           | –                      |